Reference-Guided Merge Tree Maps: Quantifying Position–Topology Trade-offs

Category: Research

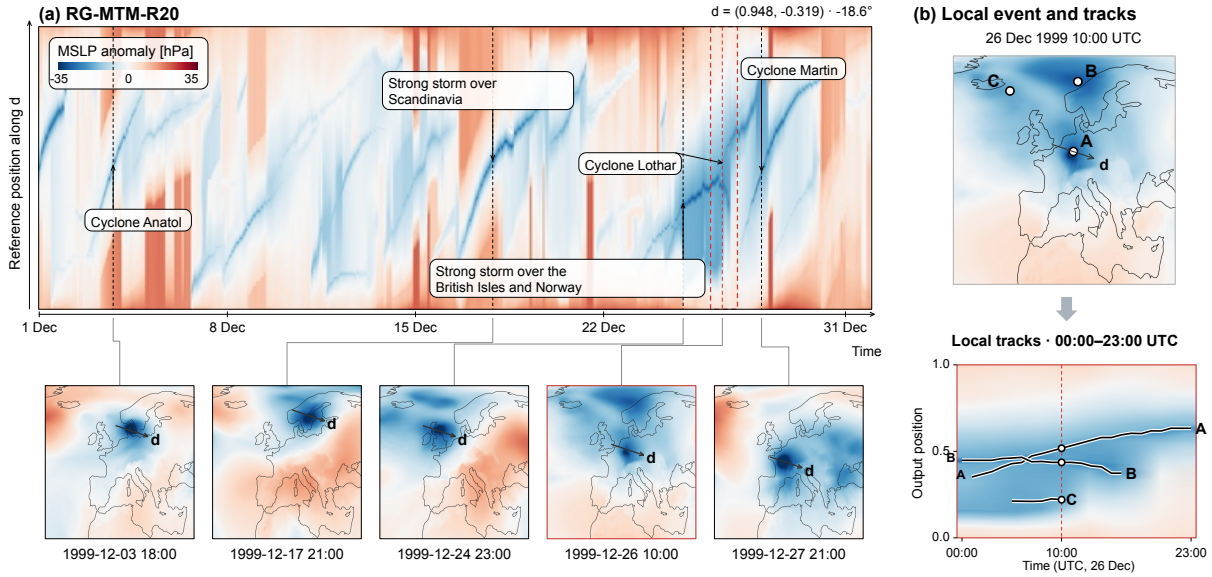


Figure 1: RG-MTM-R20 visualization of hourly Storms pressure anomalies over December 1999. (a) All 744 hourly fields are mapped along the data-derived fixed direction $d \approx (0.948, -0.319)$ and linked to five source snapshots. Blue and red encode negative and positive anomalies on a shared $[-35, 35]$ hPa scale. Event names and times follow TMTM [19]; labels identify computed extrema near the reported regions rather than independently validated cyclone tracks. The red dashed outline selects all of 26 December. (b) The 10:00 source field marks $A = 130$, $B = 113$, and $C = 133$; the lower map retains all 24 hourly profiles over the full $[0, 1]$ output axis. Solid curves show saved anchors over 00:00–23:00 UTC, with gaps outside saved lifetimes; C exists only from 05:00 to 10:00. Anomalies subtract a centered 192-hour mean; layout uses the RG-MTM 49×49 grid and 250 km smoothing. Display interpolation preserves each hourly profile. This visualization is independent of the 118-frame Storms quantitative evaluation.

Abstract

Merge tree maps summarize scalar fields through a hierarchy of extrema and their merge relations. Keeping each subtree contiguous can displace features from their projected spatial positions. We present Reference-Guided Merge Tree Maps (RG-MTM), which diagnoses this conflict and constructs maps with an explicit position–topology trade-off. A positional certificate separates interval-packing cost from the additional displacement imposed by hierarchy. Optimal scalar filling for a fixed leaf order preserves extremum values and minimizes labelled pairwise merge-level error, yielding exact attainable frontiers for small trees. RG-MTM-P preserves source merge levels; RG-MTM-R selectively releases contiguity constraints and reports the resulting scalar cost. Across four real and two synthetic temporal sequences comprising 342 frames, both modes improve reference order over shared-input TMTM and ST-MTM baselines. On Storms, the 20% eligibility setting (R20) reduces P's order error from 3.77% to 1.16%, with 1.47% mean merge-level error; its motion-proxy errors decrease to 19/104 from ST-MTM's 43/104. Independent experiments cover Tangaroa (201 original-grid volumes), Cylinder (102 volumes), one static Nucleon slice, and 21 reconstructed Benzene potential sections. On Tangaroa, R20 reduces P's order error from 9.61% to 0.019%, at 14.44% mean merge-level error. These results quantify the reference alignment gained and scalar connectivity changed by a chosen order, while retaining the source fields for spatial interpretation.

Index Terms: Temporal scalar fields, merge tree maps, reference-guided layout, positional certificates, merge-level distortion.

1 Introduction

Time-varying scalar fields describe pressure, fire activity, radiation, pollution, and simulated flow. Their extrema identify intensity regions; merge levels record when these regions connect. Static temporal summaries expose feature strength, lifetime, and connectivity across distant time steps.

Temporal merge tree maps (TMTM) linearize each field through its augmented merge tree and align consecutive profiles through subtree overlap [19]. Spatiotemporal merge tree maps (ST-MTM) introduce a distance-aware one-dimensional skeleton and temporal coherence [8]. Both generate orders by recursively arranging complete child subtrees. This keeps hierarchical features together, but the resulting axis follows tree connectivity rather than a fixed spatial coordinate.

The distinction matters when position is part of the analysis. The frame-48 Storms case in Supplement S8 exposes this conflict: the source tree groups A, B, while C lies between their reference positions along the selected direction. Keeping A, B contiguous excludes their geographic order: at least one feature must move away from its reference position. Admitting that order instead changes the scalar barriers representing their merge relations. The choice concerns both spatial alignment and the pressure threshold at which structures connect.

We ask **RQ1**: when hierarchy must be preserved, why is reference displacement unavoidable, and how small can it be? **RQ2**:

which hierarchy constraints can be released to improve alignment, and what merge-level change does the resulting map incur? **Reference-Guided Merge Tree Maps (RG-MTM)** uses a fixed spatial projection as its target. It diagnoses unavoidable displacement before layout, then constructs either a preserving map, RG-MTM-P, or a selectively relaxed map, RG-MTM-R. Both retain the extrema and their values; the relaxed map reports the change in their pairwise merge levels.

Our evaluation also asks **RQ3**: how do these choices affect reference alignment, scalar connectivity, and motion proxies across temporal, static, and volumetric inputs?

The paper makes three contributions:

- A positional certificate that separates interval-packing and hierarchy costs, together with conflict triples that identify the obstructing feature relations.
- An optimal fixed-order barrier construction and exact small-tree position–topology frontiers, connecting a changed leaf order to its minimum scalar cost.
- A reference-guided construction connecting diagnosis, hierarchy selection, interval layout, and scalar filling, evaluated on six planar temporal sequences, an independent reference-position control, two original-grid volumetric sequences, and static Nucleon and reconstructed Benzene inputs.

The evaluation measures reference order, achieved anchor displacement, rendered merge levels, and approach/separation decisions. Source fields, temporal maps, and error records connect these measurements to the feature relations changed by each mode. Sections 3–4 establish the model and its guarantees; Section 5 describes the construction. Section 6 tests the mechanisms and shared-input comparisons, Section 7 examines source-linked results by spatial dimension, and Section 8 discusses interpretation and limits.

2 Related Work

Temporal summaries and spatial linearization. Generalized space–time cubes describe temporal visualizations through transformations of space and time [1]. Studies of animation and trend analysis show how presentation and comparison tasks affect the usefulness of animated and static displays [30, 37]. Static temporal maps place distant time steps in one view, with their interpretation determined by the compressed spatial axis. Hovmöller diagrams retain a selected spatial coordinate [15]; geographic orderings and context-based space-filling curves construct spatial traversals [7, 13]. Dynamic Volume Lines apply space-filling traversals to volumetric comparison [42], while one-dimensional projections and data-driven curves adapt the ordering to spatiotemporal fields [12, 47]. MotionRugs instead organize moving point features [5]. RG-MTM gives the axis a fixed projection target and quantifies the displacement introduced by hierarchical feature grouping.

Feature tracking and evolving hierarchies. Feature tracking connects spatial structures across time, including evolving regions and turbulent three-dimensional features [32, 33]. Topology-based approaches organize such correspondences through multi-scale structures, global scalar-feature matching, and time-varying merge trees [4, 27, 31]. Dynamic tracking graphs support exploration of temporal relationships [44]; nested and dynamic nested tracking graphs add the hierarchical relationships of scalar features [23, 24]. Temporal Treemaps summarize evolving trees as time-aligned bands [18], and Fuzzy Contour Trees align multiple trees in a joint layout [22]. These methods establish feature identity and hierarchical context across observations. RG-MTM uses support-overlap matches as input and addresses the subsequent placement of matched extrema against a fixed reference.

Topology-based scalar maps. Topological landscapes represent scalar connectivity through terrain, with geometry-preserving extensions introducing proximity between features [3, 40]. TMTM traverses augmented merge trees to keep features intact and optimizes subtree overlap between adjacent time steps [19]. ST-MTM separates a hierarchy-constrained skeleton from leaf-arc and path filling, improving geometric distances and temporal coherence [8]. Section 3.1 and Fig. 2 summarize both workflows before defining our reference-guided extension. RG-MTM reuses this scalar/path filling and separates the original tree, which supplies merge targets, from the ordering tree, which specifies admissible permutations. Pairwise distances can remain unchanged under a global translation or reflection, whereas fixed-reference residuals detect either transformation. RG-MTM therefore adds an absolute positional target to the geometry-aware design space. Its certificate measures the reference-position obstruction, and its scalar correction minimizes merge-level error for the selected order and fixed extremum values.

Ordering, simplification, and tree comparison. Optimal leaf ordering searches tree-consistent permutations [2]. Separation-constrained layout and interval separation control overlap and displacement [10, 21]. RG-MTM couples these constraints with reference coordinates, interval widths, and anchor slack. Persistence simplification removes scalar features [11, 36]; our ordering relaxation retains the leaves and releases selected contiguity constraints. Topological comparison methods span several descriptors and matching assumptions [46], including labelled cophenetic/interleaving comparisons [25, 45], Wasserstein merge-tree comparisons [28], and branch-decomposition-independent edit distances [43]. Our discrepancy uses fixed leaf correspondences to measure the scalar change introduced by an order and its filled profile.

Layout quality. Stress and trustworthiness assess distance and neighbourhood preservation [26, 34, 38]; ST-MTM also evaluates temporal displacement [8]. We combine these measures with reference-order error, actual anchor displacement, and rendered merge-level error, distinguishing the spatial, temporal, and topological properties represented by the map.

3 Background and Problem Formulation

3.1 From Augmented Trees to Temporal Maps

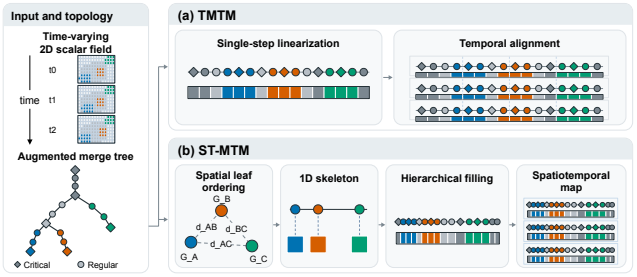


Figure 2: Conceptual comparison of preceding merge-tree map workflows, redrawn and extended from ST-MTM Figs. 1 and 3 [8]. Both use augmented trees with critical (diamond) and regular (circle) nodes. (a) TMTM [19] linearizes samples within subtrees and optimizes temporal alignment. (b) ST-MTM constructs a distance-aware 1D skeleton, fills its scalar domain, and assembles the map. Branch/sample identities use the same palette throughout the schematics: A (blue), B (orange), and C (green); internal groups are grey.

Figure 2 summarizes the two constructions on which RG-MTM builds. Both start from an augmented merge tree for each scalar-field observation. Critical nodes describe extrema and component

merges; regular nodes retain samples along the arcs. The branching hierarchy groups connected scalar regions, while augmentation carries their values into the linearized field.

In TMTM, a recursive traversal maps all tree samples to a one-dimensional profile [19]. Each subtree occupies a contiguous interval whose length is proportional to its sample count. Changing child order changes the linearization while preserving the merge tree. Temporal optimization selects these branch decisions to make interval overlaps between adjacent profiles agree with source-space subtree overlaps; stacking the aligned profiles produces the temporal map (Fig. 2(a)).

ST-MTM separates geometric layout from scalar filling [8]. It first reorders leaves within the hierarchy using distances between feature centroids, then optimizes a one-dimensional skeleton to approximate feature distances under non-overlap constraints. Its temporal mode also uses tracked leaves to improve consistency between consecutive layouts. Leaf-arc sampling and hierarchical path filling populate the resulting intervals and gaps with scalar values before temporal assembly (Fig. 2(b)). The diagram's lower path summarizes these stages; its temporal optimization acts during skeleton construction.

These constructions establish topology-preserving linearization and geometry-aware placement, respectively. Their admissible orders keep each subtree contiguous. A fixed spatial projection can instead request interleaved leaves from different subtrees, making reference alignment incompatible with that order constraint. RG-MTM diagnoses the resulting displacement, selects which constraints to retain or release, and measures the scalar cost of the chosen order. It reuses the augmented trees and scalar/path-filling framework while adding reference coordinates, positional certificates, and optimal barrier correction.

3.2 Scalar Features and Merge Relations

The input comprises processed fields $F_t : \Omega \to \mathbb{R}$, augmented merge trees T_t, leaf supports, and temporal matches. For a join tree, leaves are minima and internal nodes record the levels at which sublevel-set components merge. Split trees follow the same convention after negating scalar values. Augmentation retains regular samples along tree arcs, providing field context for the one-dimensional profile.

Leaf i has extremum position e_i^t, value $f_t(i)$, support measure μ_i^t, and support centroid G_i^t. The measure is area for a planar field and volume for a volumetric field. The extremum defines the positional reference; the centroid supplies the geometric distance term. Adjacent-frame support intersections define a maximum-weight Hungarian matching, with positive-overlap assignments continued as tracks.

Contracting unary chains gives the branching hierarchy for ordering. A legal permutation $\pi \in \Pi(T)$ keeps every subtree's leaves contiguous. For a join profile ψ, the merge level of two labelled extrema is the maximum on their inter-anchor interval I_{ij}. Its discrepancy from the source tree is

$$d_{\text{top}}(\psi, T) = \max_{i<j} \left| \max_{x \in I_{ij}} \psi(x) - f(\text{lca}_T(i,j)) \right|. \tag{1}$$

The original tree T defines these targets; the ordering tree T' defines admissible permutations; the filled profile ψ is the measured output. A single-leaf frame has zero pairwise error.

3.3 Reference Positions and Interval Geometry

A fixed projection assigns $q_i^t = \langle e_i^t, d \rangle$ to one canvas $[a, a+L]$. Each feature receives a width $w_i = c\mu_i$, interval centre z_i, and extremum anchor u_i. Its eccentricity radius is $h_i = \rho w_i/2$. The interval must lie inside the canvas, consecutive intervals must leave a gap of at least g, and the anchor must remain within h_i of its centre. Together these rules define the interval geometry $\mathscr{G}$ used by Algorithm 1 and the layout in Section 5.

The width scale c, gap $g > 0$, slack fraction $0 \le \rho \le 1$, and canvas remain fixed across time. Packing requires $\sum_i w_i + (n-1)g \le L$. Positive gaps separate the ordered anchors while allowing motion within their allocated intervals. Supplement S1 gives the complete linear constraints.

The primary position objective minimizes $\max_i |u_i - q_i|$. The scalar objective minimizes Equation (1) with fixed extremum values. Temporal layout follows the projected displacement of matched extrema. These objectives connect source coordinates, scalar connectivity, and temporal evolution within one explicit display model.

4 Position–Topology Analysis

4.1 An Attained Position Certificate

For order π, let $\tau(\pi)$ be the minimum maximum anchor displacement under the interval geometry $\mathscr{G}$. A trial budget τ gives each centre a window of radius $\tau + h_i$ around q_i, clipped to the canvas. Place intervals from left to right at their earliest feasible centres, reserving the prescribed gaps; a placement beyond an upper window makes that budget infeasible. Delaying an earlier interval cannot rescue a later upper-window violation. This test yields an attained minimum, computed directly from pairwise separation and boundary bounds by IntervalCost. Supplement S1 gives the closed form and proof; Diagnose in Algorithm 1 uses these order costs to obtain certificates.

Minimizing over tree-consistent and unrestricted orders separates the two sources of displacement:

$$\begin{aligned} \tau^*(T) &= \min_{\pi \in \Pi(T)} \tau(\pi), \qquad \tau_{\text{free}} = \min_{\pi \in S_n} \tau(\pi), \\ H &= \tau^*(T) - \tau_{\text{free}} \ge 0. \end{aligned} \tag{2}$$

The hierarchy-free optimum τ_{free} measures packing and boundary cost; H measures the extra displacement imposed by hierarchy under the same geometry. Flattening an internal branching node releases a contiguity constraint and can only decrease the optimum.

4.2 Explaining a Conflict

A conflict triple contains two leaves i, j of one subtree and an outside leaf k with $q_i < q_k < q_j$. Contiguity places k before or after the complete subtree, reversing one reference relation and forcing displacement of at least $\frac{1}{2}\min(q_k - q_i, q_j - q_k)$ in the point model. The complete certificate also incorporates interval widths, boundaries, and interacting triples.

For example, let $T = ((A,B),C)$, references $(q_A, q_B, q_C) = (20, 100, 60)$, canvas $[0, 120]$, widths 8, gap 2, and $\rho = 0.5$. The desired order A, C, B is excluded. The best tree-consistent point order needs maximum displacement 20; the full interval certificate is 23. The unrestricted interval order has zero error, giving $H = 23$. This example distinguishes order incompatibility from the additional displacement introduced by the display geometry.

4.3 Optimal Scalar Cost for a Fixed Order

If a non-root subtree v becomes non-contiguous, every line profile incurs $d_{\text{top}} \ge \gamma_v/2$, where $\gamma_v = f(p(v)) - f(v)$ is its parent merge gap. An outside leaf inserted between two subtree leaves creates an enclosing interval whose source merge requirements differ by at least γ_v. One of those requirements must change by half that gap.

OptimalBarriers in Algorithm 1 computes the exact cost $\delta^*(\pi)$ with fixed extremum values. For every gap, it first finds the smallest original merge level among leaf pairs spanning that gap. These levels C_k are the tightest barrier ceilings at zero error. It then increases a common allowance δ to accommodate the adjacent extremum values and every pair's required merge level. In the non-strict extremum model, barriers $C_k + \delta$ attain the minimum: each lies below every spanning pair's upper limit, and their interval

maxima meet all lower limits. Positive clearance retains strict extrema in the rendered implementation; Supplement S1 derives the complete expression.

In the three-leaf example, assign leaf values 0, an A/B merge at 1, and the root merge at 3. For order A, C, B, both optimal barriers equal 2 and $\delta^* = 1$. The improved reference order therefore has an explicit scalar price.

4.4 Attainable Trade-offs

Coordinates and barriers can realize their fixed-order optima together in the non-strict model; positive-clearance filling approaches the scalar optimum while retaining strict extrema. To compute the frontier at scalar budget δ, enumerate leaf orders, retain those whose OPTIMALBARRIERS cost does not exceed δ, and take their smallest INTERVALCOST. Denote this value by $F_T(\delta)$. For strictly separated branching levels, $F_T(0) = \tau^*$ and $F_T(\infty) = \tau_{\text{free}}$. A necessary lower frontier Φ_T follows by flattening nodes with $\gamma_v \le 2\delta$ and recomputing the certificate. Fixed leaf values and interacting constraints determine the gap between Φ_T and F_T. Exact enumeration supplies the small-tree reference frontiers evaluated below.

5 REFERENCE-GUIDED MERGE TREE MAPS

Figure 3 follows the data from scalar-field input through topology and projection, certificate-driven hierarchy choice, interval layout with optimal filling, and temporal assembly. The construction uses the source tree for scalar targets throughout; the ordering tree controls the permutations available to the layout. Algorithm 1 collects the construction and its hierarchy-selection and barrier routines in one procedure. Its inputs are augmented trees and tracked features from the extraction stage; its output records both the map and the achieved errors.

5.1 Shared Reference and Diagnosis

SHAREDREFERENCE derives one sequence-wide direction from consecutive-frame matches. It accumulates the outer products of matched extremum displacements, takes the leading unit eigenvector, and orients its sign toward the total displacement. Projecting retained extrema and domain corners then fixes the reference coordinates and canvas for all columns. A supplied direction supports a selected geographic axis. Feature widths use area or volume with the same conversion factor over the sequence.

DIAGNOSE assigns each frame position costs, hierarchy cost, and conflict triples. Exact small-tree evaluation returns $\tau^*(T_t)$, $\tau_{\text{free},t}$, and H_t; a bounded large-tree evaluation retains lower and upper endpoints for both position costs. Its nonnegative hierarchy lower bound is $\underline{H}_t = \max(0, \underline{\tau}^*(T_t) - \overline{\tau}_{\text{free},t})$, which coincides with H_t when both costs are exact. The gate uses this lower bound; layout uses a feasible positional upper bound. Conflict triples identify which feature relations exclude the desired order.

5.2 Preserving and Relaxed Ordering Trees

RG-MTM-P sets $T_t' = T_t$. RG-MTM-R activates when $\underline{H}_t > \theta$ and marks non-root branching nodes with $\gamma_v \le \kappa$ as eligible. Flattening these nodes joins their child blocks to their parent and permits additional interleavings while retaining all leaves. The original merge targets remain in T_t.

For the two-dimensional suite, restoration removes unnecessary flattenings. Compute the grid-certificate score b_0 after flattening every eligible node. Visit nodes in decreasing merge-gap order and restore a constraint whenever its score stays within half a pixel of the fixed b_0. The resulting tree retains the original constraints that fit the initial relaxed position score. Volumetric construction retains the eligible flattenings and uses the large-tree search described below.

Shared settings use $c|\Omega| = L/2$, $\rho = 0.5$, $g = L/240$, $\beta = L/120$, and $\theta = 0.02L$, with domain area or volume for $|\Omega|$. The main

Algorithm 1 Reference-guided temporal merge tree maps.

Data: Augmented trees T_t, extrema, supports, matches, domain Ω; mode P/R and parameters $\theta, \kappa, \beta, \mathscr{G}$
Result: Temporal map $\mathscr{V}$; position bounds, anchor residuals, and measured merge errors

Function BUILDMAP($\{T_t\}$, matches, mode)
  $(d, q, [a, a+L]) \leftarrow$ SHAREDREFERENCE(extrema, matches, Ω)
  for each time step t **do**
    Set support widths and check packing on $[a, a+L]$
    $(\mathscr{D}_t, \underline{H}_t) \leftarrow$ DIAGNOSE($T_t, q_t, \mathscr{G}_t$)
    $T_t' \leftarrow$ CHOOSETREE($T_t, \underline{H}_t$, mode)
    $\overline{\tau}_t' \leftarrow$ feasible certificate upper bound for T_t'
    $B_t \leftarrow \overline{\tau}_t' + \beta$
    $\mathscr{C}_t \leftarrow$ budget-feasible legal orders of T_t': enumerate for small trees; at most four candidates for large trees
    $(\pi, z, u) \leftarrow$ best feasible layout from candidate QPs, using the preceding frame's matched layout
    $(v_t, \alpha) \leftarrow$ RASTERIZEANDFILL(T_t, π, z, u)
    if $T_t' \neq T_t$ **then**
      $(b, \delta^*) \leftarrow$ OPTIMALBARRIERS(π, T_t)
      Correct v_t to barriers b with strict-extremum clearance; retain anchor values at α
    Record $\mathscr{D}_t$, anchor residuals, and merge error measured from v_t against T_t
  return columns $[v_1 \ \cdots \ v_K]$ with shared scalar colors

Function CHOOSETREE($T, \underline{H}$, mode)
  if mode P or $\underline{H} \le \theta$ **then**
    return T
  $W \leftarrow$ non-root branching nodes with $\gamma_v \le \kappa$
  $T' \leftarrow \text{flat}(T, W)$
  if planar protocol **then**
    $b_0 \leftarrow$ grid-certificate score of T'
    Restore constraints in T' by decreasing γ_v while its score stays within half a pixel of b_0
  return T'

Function OPTIMALBARRIERS(π, T)
  if fewer than two leaves **then**
    return no barriers, 0
  Renumber leaves by π; obtain original LCA levels M_{ij}
  $C_k \leftarrow \min_{i \le k < j} M_{ij}$ for each inter-anchor gap k
  $\delta \leftarrow \max(0, \max_k[\max(f_k, f_{k+1}) - C_k])$
  for each leaf pair $i < j$ **do**
    $\delta \leftarrow \max\big(\delta, [M_{ij} - \max_{i \le k < j} C_k]/2\big)$
  return $C + \delta$, δ

planar temporal setting is $\kappa = 0.2\Delta_F$, where Δ_F is the fixed processed scalar range. We abbreviate eligibility fractions 2%, 20%, and 100% as R2, R20, and R100. This threshold selects eligible parent gaps; it does not cap the rendered merge-level error. OPTIMALBARRIERS computes the selected order's scalar cost. The static and volumetric protocols retain their declared simplification, candidate-search, and relaxation settings (Supplement S6 and S9).

5.3 Budgeted Interval Layout

The position budget $B_t = \overline{\tau}_t' + \beta$ uses a feasible positional upper bound and reserves allowance for geometric and temporal objectives; exact diagnosis gives $\overline{\tau}_t' = \tau^*(T_t')$. For each budget-feasible legal order, a convex quadratic layout combines four mean squared residuals: anchor-reference offsets (weight 4), ordered anchor distances versus support-centroid distances (1), matched projected-motion residuals (0.5), and anchor eccentricities (0.1). The motion residual subtracts the projected source displacement from the matched anchor displacement. The interval geometry and $|u_i - q_i| \le B_t$ constrain the coordinates; the smallest evaluated objective selects the layout sequentially through time.

Small-tree construction enumerates legal orders and evaluates their continuous position costs. Large-tree construction propagates continuous earliest-finish Pareto functions through low-

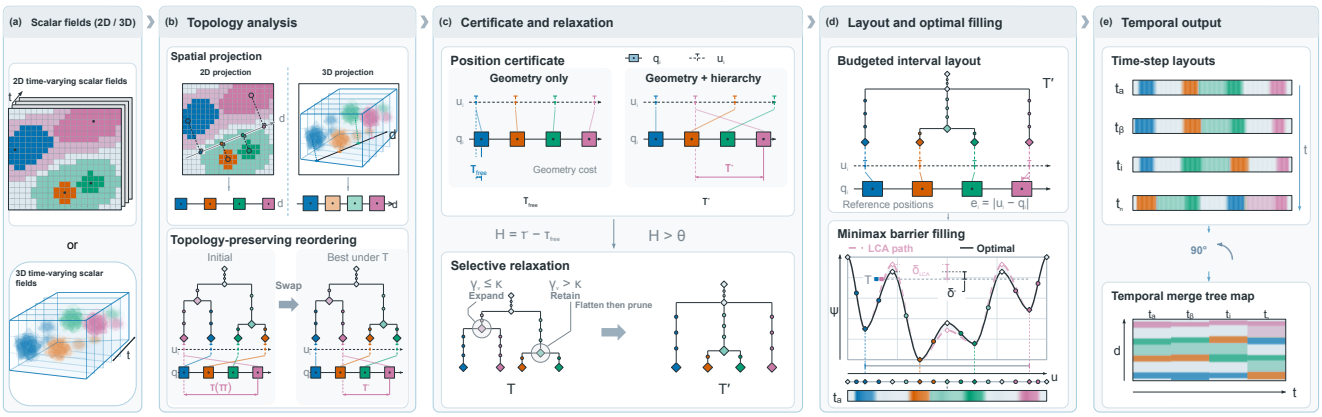


Figure 3: RG-MTM's five stages (a)–(e), using the supplied four-feature schematic. Trees and a fixed projection define source merge targets and reference positions. Certificates diagnose hierarchy-dependent displacement; selective relaxation expands the ordering tree. Budgeted intervals and minimax filling construct profiles, stacked through time. The four branch identities retain blue, orange, green, and purple across stages; actual maps encode scalar values. Flatten-then-prune restoration is used in the planar protocol. κ controls eligibility; output barriers determine measured cost.

degree nodes; higher-degree cases use workload bounds or a time-limited mixed-integer program. Each result retains a lower bound, a feasible upper bound, and a witnessing order. Layout uses the upper bound plus β and evaluates up to four legal candidates assembled from the certificate, reference-preferred, temporal-preferred, and TMTM orders. Fixed-order convex QPs record feasibility and first-order objective gaps. Figure 3 summarizes the construction; Supplement S1 details the planar construction and S6 the volumetric solver protocol.

Guarantees and selection. For a fixed order, the position cost and non-strict scalar cost are exact quantities of the display model. Exact small-tree enumeration also resolves the attainable frontier. The constructed map instead selects a secondary objective within a feasible position budget; a finite large-tree candidate set does not establish global optimality over all legal orders. Certificate intervals disclose unresolved position costs, and measured anchor and barrier errors describe the actual output. This distinction applies equally to preserving and relaxed modes.

5.4 Scalar Filling and Temporal Assembly

Scalar filling reuses ST-MTM's leaf-arc and hierarchical-path construction [8]. True extremum values occupy the anchors; resampled arc values alternate around them. Inter-anchor gaps follow original-tree LCA paths, allocating sibling, direct–deep, and deep–deep cases according to their arc lengths. Root paths fill the borders; Supplement S2 details each case. RASTERIZEANDFILL denotes this initial rendering on the shared canvas. OPTIMALBARRIERS uses join-tree values; split-tree values are negated before correction and transformed back afterwards. Its returned δ^* is the analytical fixed-order cost; measured raster error is recorded after clearance and filling.

RG-MTM-P retains the original merge barriers. For a relaxed order, call OPTIMALBARRIERS, clip each inter-anchor segment at its returned barrier, and raise its peak to that level when needed. Endpoint extrema retain their values. Figure 4 separates the order change from its scalar correction. The source tree $((A,B),C)$ excludes reference order A,C,B. For that relaxed order, original-LCA filling sets both inter-anchor barriers to s_2, changing the A/B merge by $\gamma = s_2 - s_1$. Optimal filling balances the three pairwise discrepancies at $\gamma/2$ by placing both barriers at $(s_1+s_2)/2$, provided the fixed leaf values lie below that midpoint. The dashed grey preserving profile shows the anchor displacement removed by the new order; the purple and green curves isolate filling quality at the same

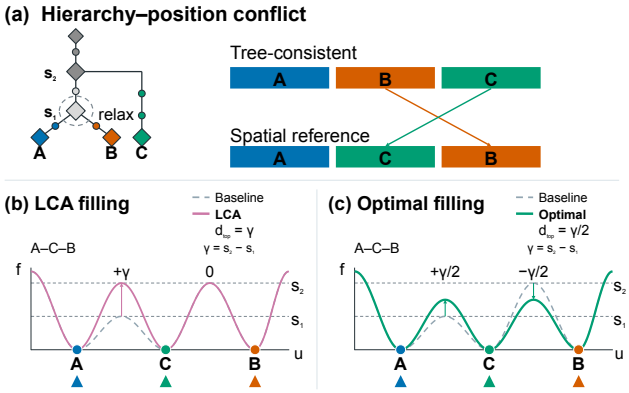


Figure 4: Hierarchy–position conflict and scalar filling for three equal-valued minima. (a) The source tree groups A,B, excluding reference order A,C,B. Identity colours match the other schematics: A blue, B orange, and C green. (b,c) Both fillings use the same relaxed anchors in order A,C,B, with identical reference triangles. Original-LCA filling (purple) sets both barriers to s_2, incurring $d_{\text{top}} = \gamma = s_2 - s_1$. Optimal filling (green) places both at $(s_1+s_2)/2$, attaining $\gamma/2$ when fixed leaf values lie below this level. The dashed grey baseline retains the hierarchy and its anchor displacement. Curves are schematic.

relaxed anchors.

The renderer maps the fixed canvas to the same raster coordinates at every time step and stacks columns chronologically. Shared colours encode scalar values; matched extrema form temporal tracks. Each frame records its position certificate, achieved anchor residuals, and actual merge-level error measured from the filled column. Nearest-pixel anchor rounding contributes at most $L/[2(N-1)]$ for raster length N, with capacity checked before filling.

6 QUANTITATIVE EVALUATION

We evaluate the position certificate (RQ1), the scalar price of changed orders (RQ2), and the alignment and motion effects of reference guidance (RQ3). Complete-sequence comparisons measure the shared-input methods, while controlled experiments isolate barrier correction and absolute anchoring. The main planar suite, additional static inputs, volumetric sequences, and visualization-only

examples retain separate populations throughout.

6.1 Data Sets, Protocol, and Measures

The 342-frame planar suite comprises four real and two synthetic sequences: Storms, 118 half-day North Atlantic/European MSLP fields (1999-11-17–2000-01-14) [14]; Wildfires, 30 two-day Australian $\log(1+\mathrm{FRP})$ aggregates (November–December 2019) [12]; NOAA OLR, 60 daily tropical Indo-Pacific fields (June–July 2024) [20]; PM2.5, 30 central-China fields (January 2000) [9]; Ring, 40 archived expanding-ring fields; and Gaussian, a local 64-frame variant with two fixed peaks, one moving peak, and a ridge. The 744-hour pressure-anomaly overview in Fig. 1 is a separate visualization experiment and is not pooled into these 342-frame statistics.

Dataset names follow TMTM and ST-MTM [8, 19]. Storms and Wildfires use their source families with different sampling and preprocessing; Gaussian is a local variant (Supplement S2).

The respective grids are 49^2, 64^2, 49^2, 49^2, 14^2, and 64^2, with 1–8, 6–11, 2–8, 1–5, 1–7, and 3–4 retained leaves per frame. Pressure and OLR use join trees; the remaining sequences use split trees. All methods share the processed fields, extracted extrema, supports, and overlap matches. Supplement S2 specifies data acquisition, smoothing, cropping, boundary treatment, and synthetic parameters.

Primary comparisons use TMTM and ST-MTM reimplementations under the shared-input protocol, RG-MTM-P, and RG-MTM-R at $\kappa/\Delta_F = 0.20$. ST-MTM uses temporal mode, uniform distance weights, $r = 0.95$, and temporal weight 0.5 (Ring: 1.5). The RG modes share geometry and secondary objectives. An independent 148-frame Storms/Wildfires audit matches 486 TMTM branch decisions and 1118 interval/anchor records to the official C++ core, and checks ST-MTM objectives, constraints, and rendered topology (Supplement S7). Supplemental controls comprise native Hilbert/Morton traversals, packed reference sorting with optimal barriers, fixed axes, and exact reference projection. A separate 64-frame mirror-motion experiment tests reference anchoring. Section 7.2 reports the 201-frame Tangaroa and 102-frame Cylinder experiments independently; their additional R settings and three-dimensional motion diagnostics are not pooled into the planar results.

Additional two-dimensional inputs. We additionally evaluate the original author Nucleon slice and 21 reconstructed two-dimensional Benzene potential sections (Sections 7.1.3 and 7.1.4; Supplement S9). Tables 1–2 collect all planar measurements, with the static populations kept separate from the 342-frame temporal suite. Tangaroa and Cylinder are grouped later as three-dimensional inputs in Section 7.2 and Table 3. Benzene sections represent a static point-charge model rather than physical time or the original author potential.

Reference order and actual displacement. Normalized Reference-Order Error (NROE) evaluates the displayed anchor order π_t independently of native scales and widths:

$$\mathrm{NROE}_t = \frac{\max\{0, \max_{r<s}(q^t_{\pi_r} - q^t_{\pi_s})\}}{2L}, \qquad \mathrm{NROE} = \mathrm{mean}_t\, \mathrm{NROE}_t. \tag{3}$$

It is the smallest maximum displacement after monotone calibration, with stable tie-breaking for coincident anchors; single-leaf frames have zero order error. Actual RG coordinates use $E_{\mathrm{pool}} = \sum_{t,i} |u^t_i - q^t_i| / (L \sum_t n_t)$ and $E_\infty = \max_{t,i} |u^t_i - q^t_i| / L$, computed from continuous layouts. These separate order compatibility from achieved position.

Merge levels. Normalized Merge-Level Error is $D_t = d_{\mathrm{top}}(\psi_t, T_t)/\Delta_F$, using a fixed processed scalar range for the sequence. $\mathrm{NMLE}_{\mathrm{mean}}$ averages framewise worst-pair errors; $\mathrm{NMLE}_{\mathrm{max}}$ gives the worst frame. Values in Table 1 are measured from filled-raster barriers. The analytical δ^* supplies the corresponding fixed-order prediction. This metric evaluates merge thresholds for corresponding leaves; it does not measure all changes to the scalar distribution or thresholded spatial regions.

Motion and geometry. A tracked pair supplies a clear approach/separation event when its spatial distance changes by more than 2% of domain diagonal across two steps. The displayed decision uses anchor-distance change and a 2% threshold of the method's sequence-wide occupied range. Motion error ME counts reversals plus misses among shared N_{clear} events. Four-step circular temporal blocks give paired bootstrap intervals. Support-centroid scale-normalized stress (SNS), trustworthiness (TW), and full-canvas temporal displacement (TD) quantify additional geometric properties; their definitions and complete results are in Supplement S3. Motion counts are geometric diagnostics of overlap-linked extrema. Small event populations and a zero-event case are reported explicitly, and no aggregate accuracy is inferred from averaging datasets with different event counts.

6.2 Position Certificates and Scalar Correction

Hierarchy cost exceeds 2% of reference extent in 40/118 Storms, 19/30 Wildfires, 7/60 OLR, 5/30 PM2.5, 21/40 Ring, and 47/64 Gaussian frames. Hierarchy contributes more than packing in every sequence: respective mean costs are 4.17%/0.16% on Storms and 6.12%/0.06% on Wildfires; Supplement S3 retains all six pairs.

Independent LP and order-enumeration checks connect the position formulas to feasible layouts. Measured barriers agree with the optimal filling to within 1.2×10^{-5} of the processed scalar range, including the strict-extremum clearance. TMTM, ST-MTM, and RG-MTM-P have zero measured labelled pairwise merge error across all suite frames. For the selected RG-MTM-R orders, fixed-order position optima reach the exact frontier within half a 1024-grid pixel in 339/342 frames. Figure 5 presents the selected five-sequence position–scalar overview, including archived Storms 2013/14 as an additional example. The 339/342-frame check uses the six active sequences and each frame's own cost; it is independent of this archived panel. Frontiers use one common budget; working points combine mean positional optima with maximum scalar costs. Table 2 separately measures final anchors.

Holding the selected order fixed isolates scalar correction. Optimal barriers reduce mean Storms NMLE from 2.07% for original-LCA filling to 1.47%, and Ring from 5.46% to 2.73%. Packed reference sorting reaches zero Storms NROE with 6.18% mean NMLE, compared with RG-MTM-R's 1.47%. These controls separate the effects of reference ordering, hierarchy choice, and barrier construction.

6.3 Comparison with TMTM and ST-MTM

Within the six-sequence suite, RG-MTM-P lowers NROE relative to both baselines on every sequence while retaining the source merge levels (Table 1). Storms changes from ST-MTM's 16.28% to 3.77%, and Wildfires from 24.02% to 5.83%. This isolates the benefit of a reference-guided layout within the original hierarchy.

Relaxation improves NROE and pooled displacement on all six sequences. Storms changes from 3.77% to 1.16% NROE, 4.13% to 2.14% pooled displacement, and 26.19% to 18.46% maximum displacement. Wildfires reaches 0.34% NROE and 0.80% pooled displacement. Tables 1–2 retain every sequence's order and coordinate measurements.

For the six temporal sequences, mean relaxed NMLE ranges from 0.86% to 3.96%, with maxima of 9.48%–33.56%. Storms

Table 1: Reference-order error, motion-proxy error, and measured merge-level cost for all two-dimensional inputs. The six temporal sequences and the two static inputs are reported separately.

		TMTM		ST-MTM		RG-MTM-P (ours)		RG-MTM-R (ours)			
Data	N_{clear}	NROE↓	ME↓	NROE↓	ME↓	NROE↓	ME↓	NROE↓	ME↓	$\text{NMLE}_{\text{mean}}$ ↓	NMLE_{max} ↓
Storms	104	19.23	0.471	16.28	0.413	3.77	0.231	1.16	0.183	1.47	26.07
Wildfires	112	29.30	0.393	24.02	0.357	5.83	0.321	0.34	0.312	3.96	19.56
OLR	56	19.31	0.446	17.21	0.214	0.89	0.054	0.00	0.054	1.10	33.56
PM2.5	7	11.86	0.714	11.30	0.429	1.11	0.429	0.16	0.286	0.86	9.48
Ring	124	7.18	0.645	10.39	0.347	2.66	0.315	0.13	0.129	2.73	12.35
Gaussian	0	25.76	–	25.76	–	5.43	–	1.38	–	3.35	9.98
Static two-dimensional inputs (independent of the 342-frame suite)											
Nucleon	–	11.9048	–	n/r	–	2.3810	–	2.3810	–	0.0000	0.0000
Benzene	–	50.0000	–	n/r	–	4.2063	–	0.0000	–	0.0025	0.0080

NROE: % of reference extent; NMLE: % of processed scalar range; ME: error fraction over shared N_{clear} two-step events at 2% thresholds. – denotes inapplicable motion or no clear events; n/r means ST-MTM was not run for the static additions. Benzene is reconstructed potential over 21 static two-dimensional sections, not physical time. TMTM, ST-MTM, and RG-MTM-P have zero measured merge-level error. RG-MTM-R uses $\kappa/\Delta_F = 0.20$, a node-eligibility threshold; mean and maximum NMLE disclose its achieved cost.

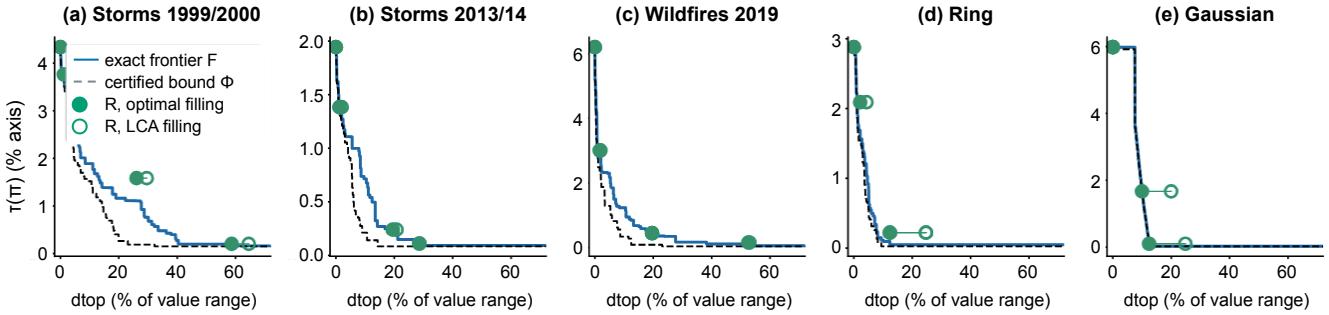


Figure 5: Position–topology trade-offs in the selected five-sequence overview. Curves show temporal means of exact per-frame frontiers F (blue) and certified lower bounds Φ (dashed) at a common scalar budget. Green points compare optimal (filled) and original-LCA (open) filling for identical selected orders at $\kappa/\Delta_F \in \{0, 0.02, 0.20, \infty\}$. Their horizontal coordinates are cross-frame maximum scalar costs; vertical coordinates are mean fixed-order position optima $\tau(\pi)$. Storms 2013/14 is an additional archived sequence, separate from the 342-frame main suite.

Table 2: Two-dimensional inputs: achieved continuous reference displacement (% of L).

	RG-MTM-P		RG-MTM-R, $\kappa = 20\%$	
Data	E_{pool}	E_∞	E_{pool}	E_∞
Storms	4.13	26.19	2.14	18.46
Wildfires	3.18	20.48	0.80	6.64
OLR	1.00	17.14	0.57	1.69
PM2.5	2.30	10.24	1.18	5.60
Ring	2.43	11.90	0.71	2.52
Gaussian	4.18	12.11	1.65	8.07
Nucleon	1.92	3.43	1.92	3.43
Benzene	1.60	15.39	0.52	1.75

E_{pool} pools retained feature-frame anchors; E_∞ is the largest continuous anchor error. Both use % of reference extent, before raster rounding. Nucleon and Benzene use saved continuous layouts and the same definition; their populations remain independent of the temporal suite.

incurs 1.47% mean and 26.07% maximum NMLE; Wildfires incurs 3.96% and 19.56%. These measured barriers quantify both sequence-wide and worst-frame costs against the source trees.

On Storms, motion-proxy errors decrease from ST-MTM's 43/104 to RG-MTM-R's 19/104 events. The paired difference is -23.1 percentage points, with a 95% temporal-block interval of $[-33.6, -12.8]$. Ring decreases from 43/124 to 16/124. The complete table preserves the shared event counts and both RG modes for every sequence.

Geometry remains a complementary criterion. On Wildfires, R reduces ST-MTM's SNS from 0.0786 to 0.0579 and TD from 7.9585 to 3.2487. Supplement S3 gives all geometric results, direction/persistence/extraction/eligibility controls, baseline settings, motion windows, and the separate large binary-tree benchmark.

6.4 Absolute Reference and Hierarchical Ordering

The mirror-motion control separates fixed-reference anchoring from pairwise-distance preservation (Fig. 6; protocol and measurements in Supplement S2, full visual comparison in S10). Across 64 fields on a 241^2 grid, B moves from $(40, 80)$ to $(80, 40)$ along the perpendicular bisector of fixed peaks $A = (20, 20)$ and $C = (100, 100)$. The mirror endpoints have identical distance matrices but different $q_i = x_i$. TMTM and ST-MTM keep B near a constant map ordinate; RG-MTM-R follows its increasing reference. Final-frame position errors are 16.67%, 16.67%, and 0.84%, using each baseline's best sequence-wide affine minimax calibration, including one global reflection. All three preserve merge levels. These 64 frames are independent of the 342-frame main suite.

Hierarchical row ordering. The certificate also diagnoses a clustering hierarchy of 1024 active wildfire cells, each containing 30 two-day observations. Optimal leaf ordering displaces some cells by 88.23% of display height, compared with the hierarchy-constrained optimum of 67.29% and the free-packing optimum of 20.55%. Their separation distinguishes the chosen ordering from the cost imposed by subtree contiguity. Supplement S10 retains the full side-by-side display and protocol; S3 gives numerical precision and records.

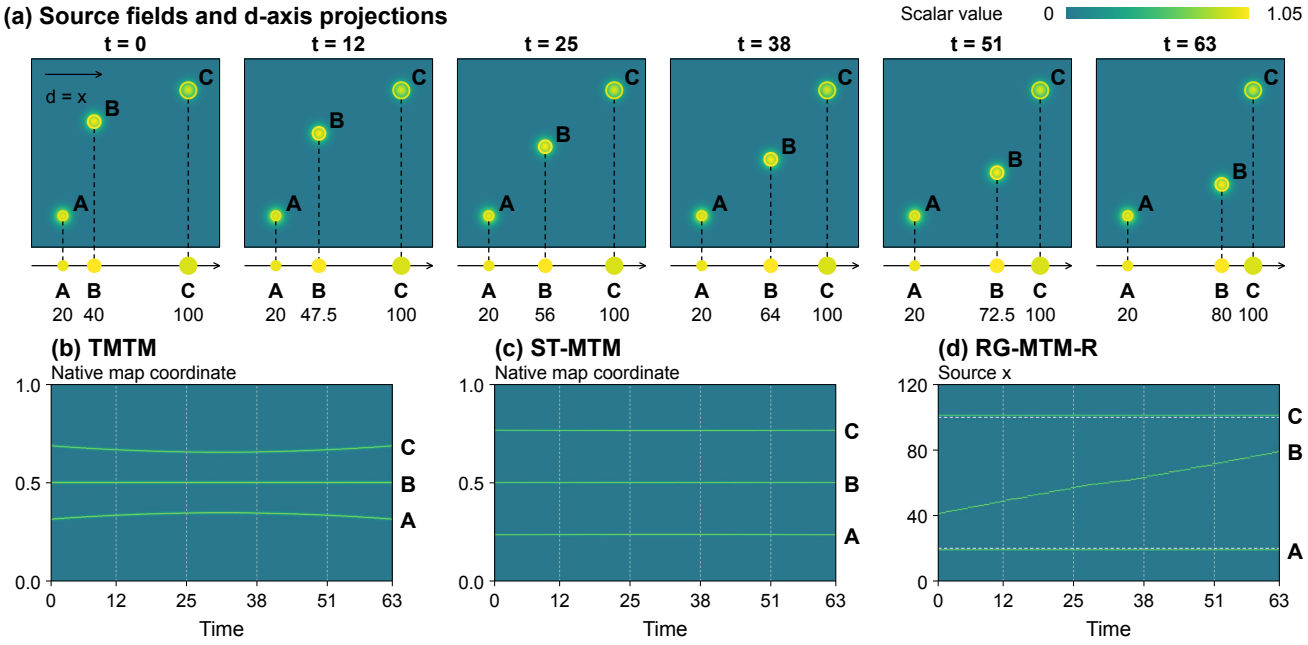


Figure 6: Relative distances and absolute reference position. (a) Six source fields and x projections follow moving B between fixed A, C; mirror endpoints preserve pairwise distances while changing horizontal position. (b,c) TMTM/ST-MTM use internal map coordinates. (d) RG-MTM-R uses source x units, with dashed fixed references. Source peaks and temporal bands share a linear 0–1.05 scalar scale. Vertical guides mark source-frame times. All three preserve merge levels; position errors and calibration are reported in Section 6.4.

6.5 Data Size and Topological Complexity

Spatial resolution determines the cost of tree extraction and the number of samples available for scalar filling; the number and branching degree of retained features determine the difficulty of positional diagnosis and order selection. Raster resolution controls how finely anchors and barriers appear in the final map. These are separate scales: a large input volume need not require exhaustive ordering of all its samples.

The retained binary-hierarchy benchmark increases from 64 leaves on 256 rows in 0.0041 s to 4096 leaves on 16384 rows in 3.604 s. These measurements concern positional diagnosis in the recorded benchmark environment. They do not include field preprocessing, scalar filling, or the quadratic layout. Exact small-tree frontiers enumerate permutations; flattening nodes can increase branching degree and the cost of subset-based diagnosis. The degree-controlled benchmark and timing details are in Supplement S3.

7 Results

The results connect the quantitative comparisons to source-space interpretation. We first examine planar temporal maps, then a static slice and a sequence of static sections, followed by volumetric temporal data. Each comparison uses the same source features in P and R and reports the scalar connectivity exchanged for improved reference alignment. Temporal overviews, scalar profiles, and source views have distinct roles: they expose evolution, show connection thresholds, and locate features in the original domain, respectively.

7.1 Two-Dimensional Data

7.1.1 Storms: Atmospheric Pressure

Figure 1 shows an additional hourly Storms visualization covering 1–31 December 1999 (744 hourly fields). It uses mean sea-level pressure anomalies relative to a centered eight-day average, with blue for negative anomalies and red for positive anomalies. The fixed direction $d \approx (0.948, -0.319)$ is estimated from the overlap-linked extrema motions and shown in each source snapshot. Five event times reported by TMTM [19] connect the temporal overview to the original hourly geographic fields. The annotated extrema provide spatial context at those times; their track identifiers are computational correspondences, not independently validated meteorological identities. This experiment retains the RG-MTM spatial processing and does not replace or extend the 118-frame half-day sequence used for the reported planar statistics. Figure 1(b) retains the complete output-position axis for all 24 hourly profiles on 26 December, with three computed minima overlaid. The full-height red dashed outline in the overview selects this calendar day; no spatial or output-position subset is selected. The saved anchors of A and B exchange relative order, while the shorter-lived C is shown only over its recorded 05:00–10:00 correspondence interval. The source markers and curves use the same track IDs (130, 113, and 133); no trajectory is continued beyond its saved lifetime.

In the separate 118-frame evaluation sequence, the 11 December 1999 case (Supplement S8) illustrates how geographic location and pressure connectivity interact. The source tree connects minima A, B before the outside minimum C joins, but the projection places C between them. A preserving layout therefore moves at least one feature away from its reference. The certificate attributes 8.45% of reference extent to this hierarchy obstruction. R admits the spatial order A, C, B and places the marked tracks close to their source references. Across all five features at this time, maximum displacement decreases from 8.87% for P to 0.83% for R. The corresponding TMTM, ST-MTM, P, and R panels are retained in Supplement S8.

The improved alignment has an interpretable scalar price. The A/B connection changes from approximately 994.1 to 1006.7 hPa, while its connections to C change from 1009.4 to 1006.7 hPa. These are the pressure thresholds at which the displayed components connect. The worst-pair change is 12.558 hPa, or 13.21% actual NMLE; the complete-sequence mean is 1.47%. Thus, the map exposes both the geographic order gained by relaxation and

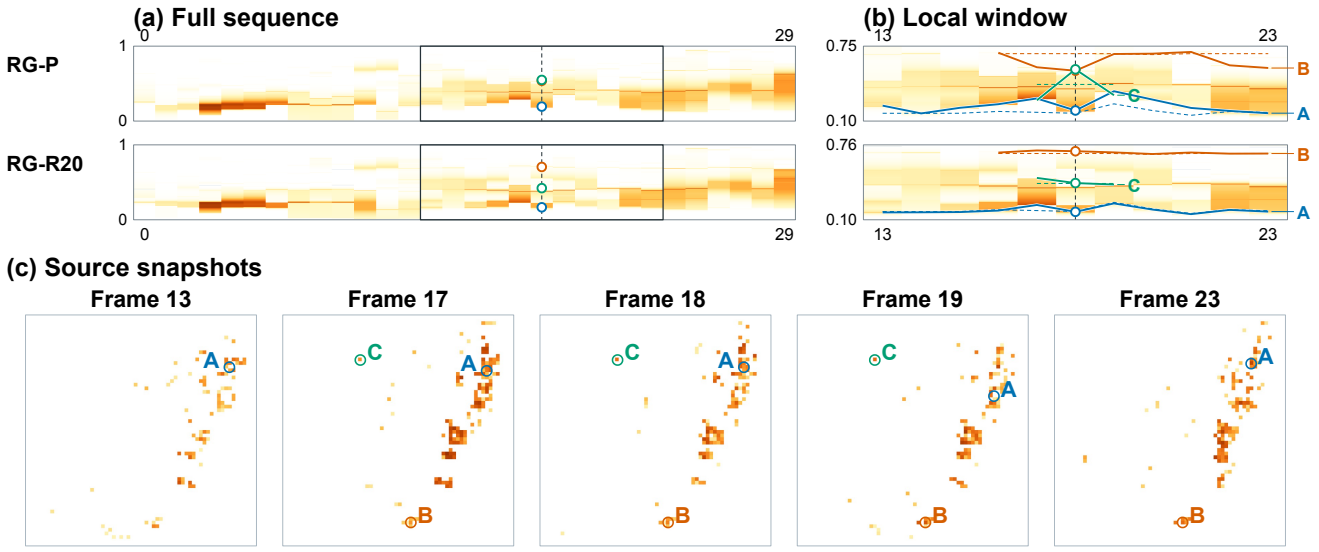


Figure 7: Wildfires: RG-MTM-P/R over all 30 frames (a), with frames 13–23 enlarged (b), and raw source snapshots at frames 13, 17, 18, 19, and 23 (c). Tracks $A = 3$, $B = 48$, $C = 52$ identify the same processed extrema. Solid traces show anchors; dashed traces show fixed references. Yellow–orange–brown encodes log-FRP, with ranges 0–4.21 for processed maps and 0–12.05 for raw fields. Grey-blue marks artificial negative connections. At frame 18, relaxation reduces maximum displacement from 14.75% to 2.63% at 15.09% actual NMLE. Complete baseline panels are in Supplement S10.

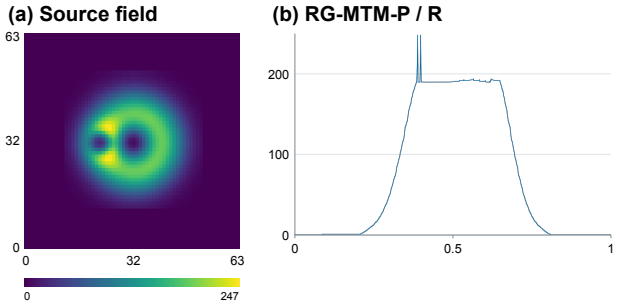


Figure 8: Nucleon: the unchanged author 64×64 scalar slice (a) and the complete RG-MTM-P profile (b, 32768 samples). Every R setting returns exactly the same profile as P. Both panels retain the original scalar units. NROE is 2.38%, with zero measured NMLE. This is one static slice; the TMTM comparison is in Supplement S10.

the pressure-connectivity detail exchanged for it. Supplement S8 provides the full frame-level measurements and analytical-versus-rendered costs.

7.1.2 Wildfires: Fire Radiative Power

The Wildfires sequence summarizes fire-radiative-power observations in 30 two-day intervals. Figure 7 shows stronger processed fire activity as orange–brown bands against the pale lower-valued background. The complete temporal maps give an overview of when these intensity patterns occur, while the source snapshots locate their spatial regions. Both RG maps use the same fixed spatial reference. Supplement S10 retains the TMTM and ST-MTM panels with their native coordinates.

Frames 17–19 illustrate why this distinction matters. The projected location of feature C stays fixed, yet its P band shifts upward at frame 18 and returns in the next observation. The R band stays close to the same reference through all three frames. The spatial snapshots and matching labels separate this layout-induced displacement from a change in the feature's observed reference position. The comparison concerns the same overlap-linked features and the same scalar observations.

At frame 18, the source hierarchy groups A, B while the reference requests A, C, B. Relaxation reduces maximum displacement from 14.75% to 2.63% of reference extent. The connecting scalar profile changes as well: R lowers the A/B barrier from 0.060800 to -0.574262 in processed log-FRP units. The grey-blue under-range colour shows this artificial connection level alongside the improved anchor alignment. The resulting actual NMLE is 15.09%; the original fire observations remain nonnegative. Exact coordinates, orders, and costs are recorded in Supplement S7–S8.

7.1.3 Nucleon: Static Scalar Structure

Nucleon tests a static layout on the TMTM authors' original padded 64×64 scalar slice [19]. Figure 8 pairs the complete source slice (a) with the complete P/R scalar profile (b). Both panels describe the same static input. The input has one frame and 13 extracted extrema; the derived tree has 26 nodes, compared with 28 in the reference table. The source samples are unchanged, and no simplification threshold is applied. TMTM has 11.90% NROE, while P has 2.38%; both have zero measured NMLE. Tables 1–2 group its order, merge-level, and displacement measurements with the other planar inputs. All three R settings return exactly the same saved profile as P, so relaxation adds no benefit in this case. This null relaxation result shows that reference guidance can improve a legal order without requiring a scalar trade-off. The example supplies a static scalar comparison; its single slice supports neither temporal nor volumetric claims.

7.1.4 Benzene: Reconstructed Electrostatic Potential

The Benzene experiment uses an explicitly reconstructed electrostatic potential, rather than the unavailable original author file. An OPLS-AA point-charge model assigns $-0.115e$ to each carbon and $+0.115e$ to each hydrogen [17]. Atomic positions come from the header of a public Inviwo molecular cube [16]; its electron-density samples are not used. We evaluate $V(r) = \sum_i q_i / \|r - R_i\|$ on 21 parallel 61×61 sections, with $x, y \in [-6, 6]$ and $z \in [-2, 2]$ in bohr. Section z is a spatial coordinate of a static field, not physical time. This classical reconstruction is a model-specific comparison, not

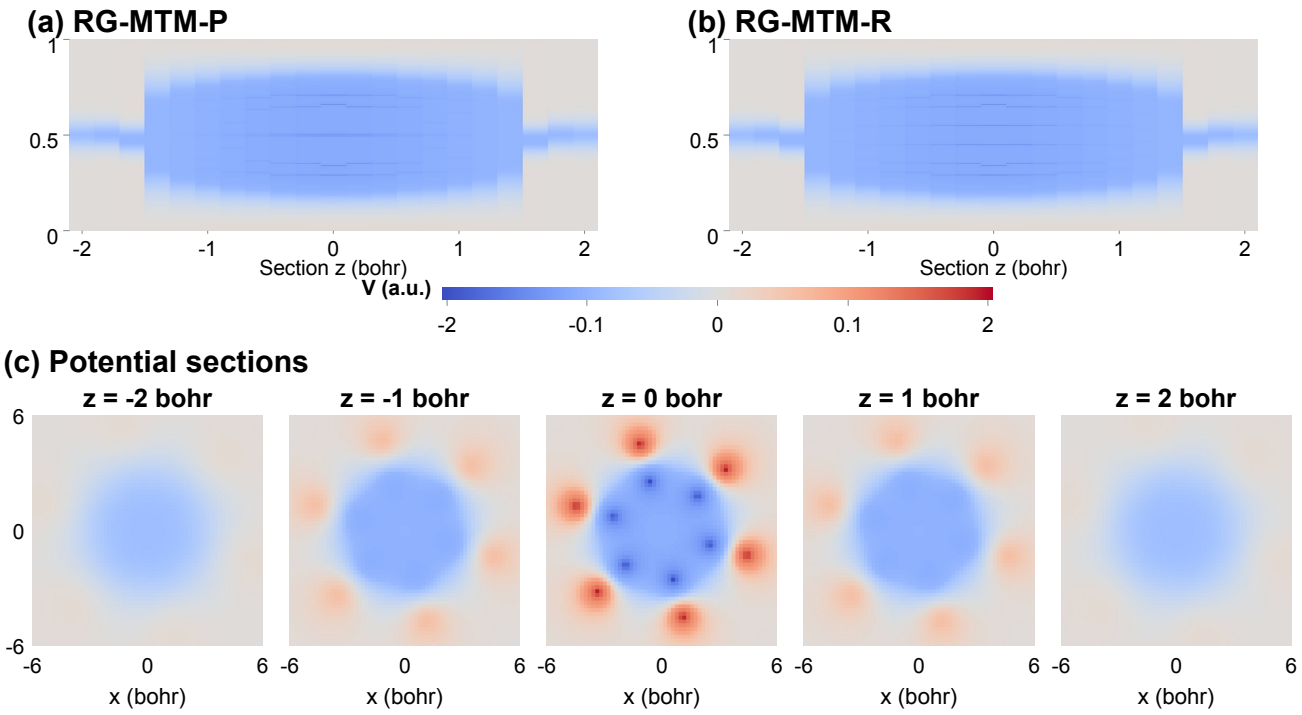


Figure 9: Benzene: RG-MTM-P/R2 maps of all 21 static z sections (a,b), with source potential at $z = -2, -1, 0, 1, 2$ bohr (c). Section z is a spatial coordinate. Maps and source fields share a symmetric-log scale, linear within ± 0.0784 a.u.; potential is in Hartree/e. This is the declared OPLS-AA reconstruction. The TMTM comparison is in Supplement S10; full and interior-extremum scores are in S9.

the original fractional-charge potential or a quantum-chemical reproduction.

Figure 9 pairs the P/R slice maps with five source sections; Supplement S10 retains the full baseline comparison. Join trees retain low-potential extrema after a common 0.05% per-section persistence simplification. P reduces mean NROE from 50.00% to 4.21% while preserving merge levels. Each R setting reaches zero NROE with 0.0025% mean and 0.0080% maximum NMLE; the three settings have identical measured results. TMTM does not optimize the supplied x reference, so these values assess reference compatibility rather than its original objective.

Four boundary extrema occur in every section and contribute substantially to the complete-leaf score. On the 15 sections with at least two interior extrema, interior-only NROE is 12.50% for TMTM, 5.89% for P, and zero for R. Across 170 overlap matches, mean adjacent-section displacement residual is 2.755% for TMTM, 1.219% for P, and 0.137% for R. Mean $\pm z$ map discrepancy similarly changes from 0.0185% to 0.0064% to 0.0030%. The source field is only approximately mirror-symmetric: its maximum paired discrepancy is 0.0241% of scalar range. These are slice-coherence diagnostics, not molecular-motion errors (Supplement S9). Keeping the boundary and full-population results together makes the scope of the apparent position gain explicit.

7.2 Three-Dimensional Data

7.2.1 Tangaroa: Velocity Magnitude

Tangaroa describes airflow around a research vessel, whose geometry influences the surrounding flow [19, 29]. Figure 10 combines temporal maps of all 201 velocity-magnitude fields with full-volume views. The maps compress the spatial domain into a fixed projected axis and expose the retained low-speed features across time; the volume views provide the three-dimensional context needed to locate those features around the vessel. All layouts use the same original-grid fields and fixed 10% persistence setting described below and in Supplement S6.

Tangaroa contains 201 velocity fields on the original $300 \times 180 \times 120$ grid, with time increment 0.01 [29]. Velocity magnitude is computed in float64 and stored as float32; magnitudes below 10^{-6} are replaced by 1 and the values clipped to $[0, 2]$. TTK 0.9.9 [35] applies traditional simplification at absolute persistence 0.2, followed by augmented join-tree extraction. This fixed 10% setting retains 18–93 leaf features per frame. We compare TMTM, P, and R at eligibility fractions 2%, 20%, and 100%. The 201-frame population is independent of the planar suite; rendered-anchor MAE averages within each frame and then over time; Supplement S6 gives the complete protocol.

The complete temporal maps show the effect of reference guidance on the arrangement of the bands. P preserves the source merge levels while reducing mean NROE from TMTM's 20.09% to 9.61%. R20 follows the reference order more closely, reaching 0.019% mean NROE and 1.065% mean rendered-anchor error. This arrangement incurs 14.44% mean and 31.02% maximum NMLE. Table 3 retains the intermediate R2 setting, R100, and the separately evaluated three-dimensional and projected motion decisions.

The snapshots at frames 139–141 connect this sequence-level change to individual source features. At frame 140, the source join tree groups A, B, whereas the reference places C between them. P orders the marked features A, B, C, while R20 uses A, C, B; the TMTM order C, A, B is shown in Supplement S10. The markers in the volume views and local temporal windows identify the same retained extrema. Maximum P/R displacement over all 27 features falls from 14.76% to 1.02% of reference extent, with 13.75% actual NMLE. The local profile changes the A/B and A/C connection levels by 0.275004 in speed units, making the price of the spatial reordering explicit.

Shared markers connect the sequence-wide reference map to source extrema and their changed connection levels. Supplement

Table 3: Three-dimensional inputs: Tangaroa (201 frames) and Cylinder (102 frames). Position and merge errors are percentages; motion columns give errors/shared clear events.

Method	NROE	Anchor MAE	NMLE mean	NMLE max	3D motion	Projected motion
Tangaroa (201 original-grid volumes)						
TMTM	20.092	–	0	0	6585/6755	9408/9631
RG-MTM-P	9.608	5.125	0	0	3697/6755	5045/9631
RG-MTM-R, $\kappa = 2\%$	8.560	4.311	3.515	11.537	3278/6755	3923/9631
RG-MTM-R, $\kappa = 20\%$	0.019	1.065	14.442	31.021	2060/6755	240/9631
RG-MTM-R, $\kappa = 100\%$	0.019	1.065	14.442	31.021	2060/6755	240/9631
Exact reference projection	0	0	–	–	2060/6755	0/9631
Cylinder (102 original-grid volumes)						
TMTM	26.617	–	0.000	0.000	2141/3196	1845/2934
RG-MTM-P	12.262	5.730	0.000	0.000	1191/3196	870/2934
RG-MTM-R, $\kappa = 2\%$	0.249	0.528	0.692	4.886	617/3196	123/2934
RG-MTM-R, $\kappa = 20\%$	0.084	0.509	2.810	14.281	605/3196	108/2934
RG-MTM-R, $\kappa = 100\%$	0.084	0.509	3.190	14.281	606/3196	107/2934
Exact reference projection	0	0	–	–	603/3196	0/2934

Anchor MAE averages normalized rendered-anchor errors within each frame, then across frames. NMLE uses measured barriers divided by the fixed processed scalar range (2 for Tangaroa, 25.8113 for Cylinder). Motion uses lag 2 and 2% thresholds; spatial and projected truth define their own clear-event sets. Exact projection supplies anchors $u = q$ without a scalar profile.

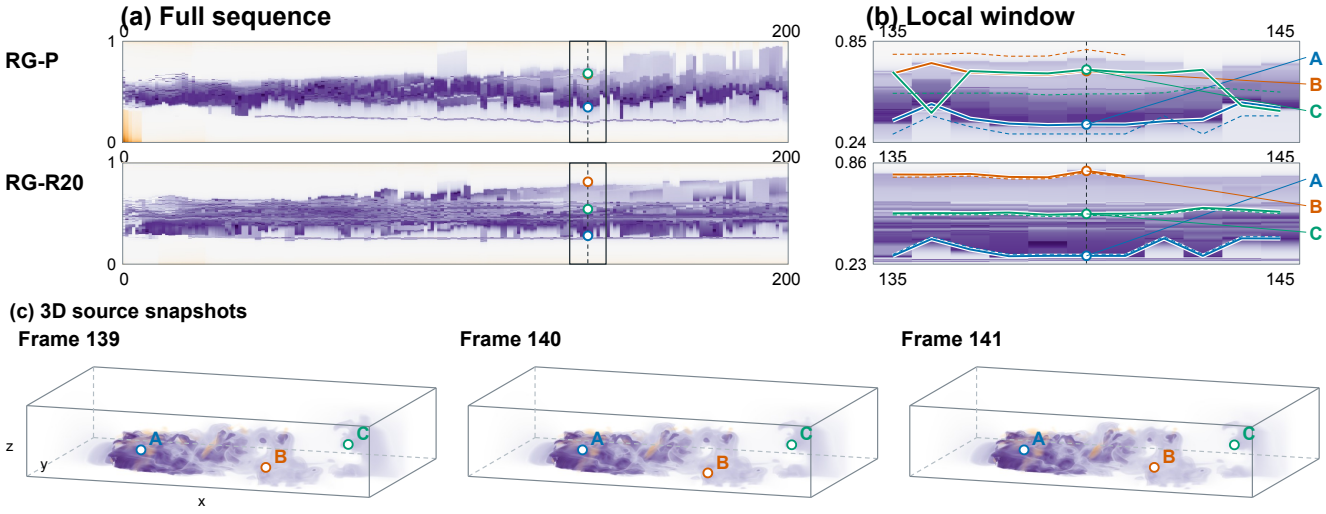


Figure 10: Tangaroa: RG-MTM-P/R20 across all 201 original-grid fields (a), the local window 135–145 (b), and full-volume views at frames 139–141 (c). Tracks $A = 432$, $B = 462$, $C = 388$ link saved anchors, dashed references, and true three-dimensional extrema. The common orthographic camera and domain frames preserve spatial context. The published purple–white–orange speed transfer function uses $[0, 2]$. At frame 140, P/R maximum displacement is 14.76%/1.02%, with 13.75% actual R NMLE. Full baseline panels and rendering details are in Supplement S6, S8, and S10.

S8 gives the coordinates and diagnostics; S6 verifies complete volumes, certificate bounds, and rendered profiles.

Tangaroa's 18–93 retained features per frame require continuous bounds and a finite candidate search rather than complete order enumeration. The large-tree solver retains certificate bounds and evaluates at most four candidate-order QPs; R2 has 48 open certificate intervals. Actual errors are measured on complete 131072-row profiles, while the 4096-row preview can omit narrow visible bands. Supplement S6 records these solver and display resolutions.

7.2.2 Cylinder: Three-Dimensional Vortex Activity

Cylinder extends the volumetric evaluation to the direct numerical square-cylinder flow of Camarri et al. [6], uniformly resampled by Weinkauf [39, 41]. All 102 public velocity volumes are used. We compute the three-dimensional $Q = \frac{1}{2}(\|\Omega\|_F^2 - \|S\|_F^2)$ on the $192 \times 64 \times 48$ velocity grid before retaining the author's $135 \times 64 \times 48$ downstream subdomain. The input is volumetric at every time, without spatial downsampling, smoothing, or feature-count selection. The 102 available times differ from the 508 reported in TMTM; this is a corresponding public-data experiment rather than a replication of its final derived input.

Figure 11 relates the P/R20 complete-sequence maps to three-dimensional source isosurfaces at frames 0, 51, and 101. A fixed x reference permits comparison of downstream feature positions. The source views show the spatial distribution of positive-Q structures, while the maps expose how their scalar patterns are arranged over the sequence. Persistence simplification is fixed at 2% of each frame's range, yielding 4–75 retained extrema; it is distinct from the RG relaxation eligibility parameter. Table 3 groups all five Cylinder configurations with Tangaroa; Supplement S9–S10 retains all five configurations and the full visual comparison.

P decreases mean NROE from 26.62% to 12.26% with zero measured NMLE. R2 reaches 0.249% NROE at 0.692% mean and 4.886% maximum NMLE; R20 reaches 0.084% NROE at 2.810% mean and 14.281% maximum NMLE. R100 has the same mean NROE as R20 but greater mean scalar cost (3.190%). Thus, wider

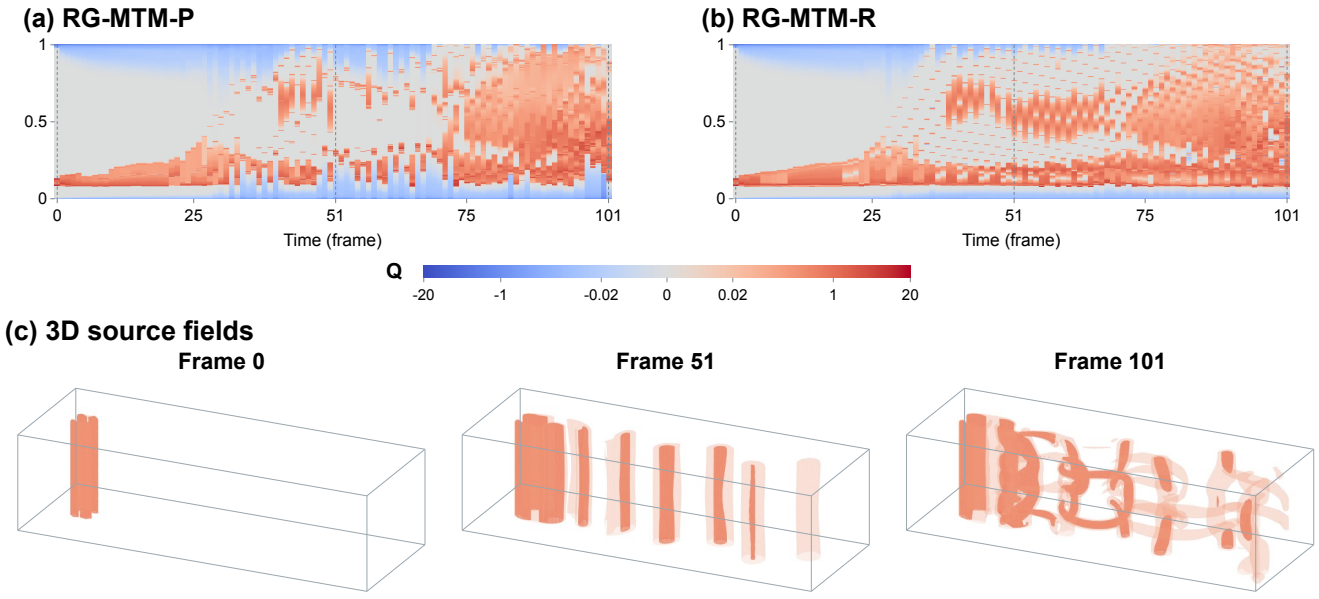


Figure 11: Cylinder: RG-MTM-P (a) and R20 (b) for all 102 original-grid three-dimensional Q fields, with source isosurfaces at frames 0, 51, and 101 (c). The maps use normalized physical x reference position; vertical guides locate the source views. Their common symmetric-log colour scale is linear within ± 0.02. Isosurfaces at $Q = 0.1$ (translucent) and 0.5 (solid) share physical aspect, camera, and domain frame. Source views precede persistence simplification; maps use the common simplified inputs. Full TMTM/R2 panels and rendering details are in Supplement S9–S10.

relaxation eligibility does not ensure a preferable measured trade-off.

For 3196 clear three-dimensional approach/separation events, errors decrease from 2141 for TMTM to 1191 for P and 605 for R20 (66.99%, 37.27%, and 18.93%). Exact x projection still makes 603 errors (18.87%), identifying spatial information lost in projection. The corresponding projected-event rates use a different denominator, 2934 events: R20 makes 108 errors (3.68%). These overlap-based geometric proxies do not establish physical vortex identities. Supplement S9 supplies the complete counts, boundary treatment, and solver records.

8 Discussion and Limitations

8.1 Interpreting Reference Alignment

The certificates isolate displacement forced by subtree contiguity (RQ1), and optimal barriers quantify the scalar price of a chosen order (RQ2). The tested reference-order gains extend across planar and volumetric inputs, while motion and geometry gains depend on the reference and event population (RQ3). Nucleon's unchanged P/R profiles show that useful alignment need not require relaxation.

A band's movement combines projected source movement and changes in its anchor-to-reference residual. Wildfires frames 17–19 show a fixed reference accompanying a shifting P anchor; matched source views separate these effects. For volumetric fields, the reference compresses two further spatial dimensions, so similar projected positions can represent features far apart in the source domain.

Reference alignment complements distance preservation and temporal coherence. On OLR, P/R motion-proxy errors remain 3/56 despite improved order, and ST-MTM retains better SNS, TW, and TD on Storms, OLR, and PM2.5. PM2.5 has seven clear events and Gaussian none at the primary thresholds. These outcomes preclude treating a lower NROE as a general guarantee of better geometry, motion interpretation, or reader performance. The motion-margin result in Supplement S1 instead certifies projected distance changes when achieved anchor errors are sufficiently small.

8.2 Choosing Preservation or Relaxation

RG-MTM-P is appropriate when source merge thresholds must remain intact. Its position certificate quantifies the displacement that the chosen geometry and hierarchy permit, while its actual anchor errors disclose the result of the budgeted secondary layout. RG-MTM-R adds reference-compatible orders while retaining the extrema and their values. Each relaxed output should be read together with its achieved displacement and mean and maximum merge-level errors; sequence means alone can conceal a large local scalar change.

The eligibility parameter κ bounds adjacent parent gaps of nodes that may be flattened. It is neither birth–death persistence nor an output-distortion tolerance. Wider eligibility need not improve the measured trade-off: Cylinder R100 has the same mean NROE as R20 but greater mean NMLE. A hard scalar-error tolerance would require constraining or rejecting candidate orders using their actual optimal filling costs and, subsequently, their rendered errors. Exact frontiers provide such a reference for small trees; the present large-tree construction uses bounded diagnosis and a finite candidate search.

Relaxation changes regular scalar values and connection thresholds. The negative Wildfires connections are artificial filling values, even though the source fire observations are nonnegative. Merge-level error describes these changed pairwise thresholds, while retaining extremum values; it does not establish preservation of scalar distributions, thresholded areas, or every internal feature. An analysis that depends on a physical threshold should consult the source field and the displayed barrier before drawing a conclusion from the relaxed map.

8.3 Scope of the Evidence

Spatial projection and preprocessing determine the structures represented by the map. Tangaroa's direction captures 57.686% of matched motion energy; exact projected anchors still produce 2060/6755 three-dimensional motion-proxy errors. Cylinder's exact x projection likewise retains 603/3196 errors. These controls

locate a limit of the chosen reference even when positional layout error vanishes. Multiple references could expose additional spatial relations, but are outside the present evaluation.

The reported populations answer different questions. The 342-frame suite compares four methods on common planar temporal inputs. The 64-frame mirror-motion control isolates absolute anchoring. Tangaroa and Cylinder test independent original-grid volumetric inputs; Nucleon tests one static slice; reconstructed Benzene tests section coherence in a declared classical potential. The 744-hour Storms overview supplies a denser visualization and is excluded from the 118-frame quantitative sequence. These populations are not combined into an overall accuracy score. ST-MTM was not run on the static additions or volumetric sequences.

Data correspondence also limits interpretation. Tangaroa includes 2459/6925 boundary-touching supports and yields 36–186 derived nodes, compared with the reference paper's 50–186. Cylinder uses the 102 public volumes and the declared non-periodic boundary treatment, whereas TMTM reports 508 frames. Benzene reconstructs a related point-charge model rather than the unavailable author potential. January PM2.5 follows a four-month exploration. These declared inputs support the reported comparisons, without establishing exact replication of every original dataset or generalization beyond the tested settings.

Overlap matches provide computational correspondences rather than validated physical tracks. Source-provider event calendars give temporal context rather than cyclone identities, and the Cylinder proxies do not establish vortex labels. Independent physical-track validation and studies of how readers interpret the maps remain necessary extensions. Further persistence settings, internal-feature evaluation, additional scalar models, and preregistered task-specific comparisons would broaden the evidence.

8.4 Computational Limits and Future Work

The closed-form fixed-order position cost and optimal non-strict barriers are exact within the stated model; complete small-tree enumeration yields exact frontiers. Large-tree certificate brackets and at most four candidate QPs provide feasible layouts with disclosed unresolved gaps. Higher-degree search and tighter certificates could reduce those gaps. A finer figure preview improves visible detail but does not strengthen the full-profile numerical evidence.

Historical planar QPs can fall back to LP-feasible layouts; complete secondary-optimality records and repeated end-to-end timings would strengthen performance evaluation. The ST-MTM reimplementation uses shared-input parameters and its SciPy OLO heuristic, which exceeds the exhaustive adjacency minimum in eight of 148 audited frames. This audit verifies the declared implementation, without establishing equivalence to every original setting.

9 Conclusion

RG-MTM makes the conflict between hierarchical feature grouping and fixed spatial reference positions explicit. Its positional certificates separate packing and boundary cost from the displacement imposed by subtree contiguity. Optimal fixed-order filling preserves extremum values and quantifies the minimum labelled merge-level cost of a changed order. The preserving and relaxed constructions turn this analysis into scalar maps whose achieved anchor and barrier errors can be inspected alongside the source fields.

The six-sequence planar evaluation, independent reference control, original-grid Tangaroa and Cylinder sequences, static Nucleon slice, and reconstructed Benzene sections show the reference alignment gained under their declared protocols and the scalar connectivity exchanged for it. The resulting maps support comparisons along a fixed reference while making the limits of projection, hierarchy, and scalar filling visible and measurable.

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